ChatGPT

4. WP2 – 2-Body Kepler + KS: Prototype Constraint-as-Moment-Map Testbed

Literature shows that the 2-body (Kepler) problem, when lifted via the Kustaanheimo–Stiefel (KS) transform, provides a clear example of an algebraic constraint functioning as a momentum map condition. In the KS regularization, one introduces a four-dimensional harmonic oscillator model whose phase space $\widetilde{M} = T^*\mathbb{R}^4$ (minus the origin) is equipped with a natural S^1 symmetry. This extra **U(1) symmetry** corresponds to the "free angle" introduced by the KS map [1]. Geometrically, $\mathbb{R}^4 \setminus \{0\}$ projects to $\mathbb{R}^3 \setminus \{0\}$ as a non-trivial principal circle bundle (the Hopf fibration) [2] [3]. The fiber S^1 is the gauge freedom in the KS transformation: rotating the 4-vector $u \in \mathbb{R}^4$ by a phase $e^{i\phi}$ does not change its image $r \in \mathbb{R}^3$. This S^1 action has an associated **momentum map** given (in suitable coordinates) by the quadratic form [4]:

$$\Xi(q,p) \;=\; q_1p_2 - q_2p_1 \;+\; q_3p_4 - q_4p_3\,,$$

which generates simultaneous rotations in the (q_1, q_2) and (q_3, q_4) planes [4]. In fact, the isotropic 4D harmonic oscillator admits a larger T^2 symmetry; one component is the total energy $H_2 = \frac{1}{2}(|p|^2 + |q|^2)$ and the other is the above S^1 phase rotation [5] [6]. **Crucially, the KS** constraints **coincide with fixing the value of this momentum map**. By enforcing the so-called "bilinear relation" of KS (which includes conditions like $|u|^2 = r$ and $\langle u, p_u \rangle = 0$), one restricts to the zero-level set (or a constant level set) of Ξ before reduction [7]. Intuitively, $\Xi = 0$ is the condition that the 4D oscillator's angular momentum in the Hopf fiber direction vanishes, so that the motion lies entirely in the reduced 3D space.

- **KS Map and Phase Space Lift:** The classical KS mapping $\Phi : \mathbb{R}^4 \to \mathbb{R}^3$ is defined by quadratic relations (originally via quaternions or spinors) that regularize the Coulomb singularity. For example, one formulation is:

  $$x = 2(u_1u_3 + u_2u_4), \qquad y = 2(u_2u_3 - u_1u_4), \qquad z = u_1^2 + u_2^2 - u_3^2 - u_4^2\,,$$

  with $r = \sqrt{x^2 + y^2 + z^2} = |u|^2$. This introduces one extra coordinate (the phase of u) beyond the three needed for $\mathbf{r} \in \mathbb{R}^3$. **The KS transform thus "lifts" the Kepler problem to an oscillator in** $\mathbb{R}^4$ [8]. The Kepler Hamiltonian $H_{\text{Kepler}} = \frac{1}{2}|p_r|^2 - \frac{\mu}{r}$ (on $T^*\mathbb{R}^3$) becomes (after time re-scaling $dt/ds = r$) the *free* harmonic oscillator Hamiltonian $H_2 = \frac{1}{2}(|p|^2 + |q|^2)$ on $T^*\mathbb{R}^4$ [8]. The price for this regularization is the extra S^1 symmetry: trajectories of the 4D harmonic oscillator correspond to one-parameter *families* of Kepler orbits. In fact, an entire **2-torus of periodic oscillator solutions corresponds to a single physical Kepler orbit** (all related by the gauge angle) [7].

- **S^1 Action and Momentum Map:** The extra U(1) symmetry in the lifted system acts by $u \mapsto e^{i\phi}u$ (rotating all four coordinates in phase). This is a **Hamiltonian S^1 action** on $T^*(\mathbb{R}^4 \setminus \{0\})$. Its momentum map $\Psi(u, p_u) = \Xi(q,p)$ is given by the conserved quantity Ξ above [4]. Setting Ξ to a constant (e.g. zero) imposes the principal **KS constraint** that links the 4D and 3D variables. In essence, the familiar KS condition $|u|^2 = r$ emerges from the geometry of the Hopf fibration – it

ensures we are restricting to one representant of each S^1 orbit. Van der Meer (2021) shows that performing **orbit space reduction by this S^1** (i.e. taking the quotient of the level set $\Xi=\text{const}$ by S^1 action) yields a 6-dimensional manifold that can be identified with the **physical Kepler phase space** [7]. This reduced space can be described concretely by a set of invariant coordinates satisfying one constraint: for instance, it is diffeomorphic to

$$T^+S^3 = \{(u,v) \in \mathbb{R}^4 \times \mathbb{R}^4 : |u| = 1,\ \langle u, v\rangle = 0,\ v \neq 0\}\,, \ =$$

enforcing a unit-norm for u and orthogonality between u and its conjugate momentum v [9]. These are precisely the **bilinear relations** that define the classical KS map in phase space.
Conclusion: the 4D oscillator's S^1 momentum map provides the **algebraic constraint** needed to descend to the 3D Kepler dynamics.

- **Symplectic Reduction and Monopole Curvature:** After imposing $\Xi=0$ and quotienting by S^1, the resulting symplectic form on the 6D reduced space differs from the naive ω_{Kepler} by a **magnetic term**. In bundle language, the lift $\mathbb{R}^4\to \mathbb{R}^3$ carries a natural **connection 1-form** (the Hopf connection) whose curvature is the area 2-form on the unit 2-sphere [10]. This curvature manifests in the reduced phase space as a **Dirac monopole field** on $\mathbb{R}^3\setminus{0}$. In practical terms, the symplectic 2-form after reduction is

  $$\omega_{\mathrm{red}} = \omega_{\mathrm{phys}} - \mu_0\, F_{\mathrm{monopole}}\ ,$$

  where ω_{phys} is the standard symplectic form in spherical coordinates and F_{monopole} is the curvature 2-form of the mechanical connection (proportional to the solid angle form on S^2). The constant μ_0 is the momentum-map level (e.g. $\Xi=0$) chosen for reduction – effectively the "charge" of the monopole field. **This is the origin of the famous magnetic term in the post-KS Kepler dynamics:** the reduced system on $\mathbb{R}^3$ can be viewed as a charged particle moving in a fictitious monopole field. Notably, the Hopf fibration provides a *global* potential 1-form on S^3 for the monopole field, even though no single-valued potential exists down on S^2 or $\mathbb{R}^3\setminus{0}$ [11] [10]. In other words, the **non-trivial bundle structure is exactly what "stores" the extra integral** (the component of the Runge–Lenz vector) that would be lost if one works only in $\mathbb{R}^3$.

In summary, the KS regularization of the 2-body problem fits perfectly into the moment-map reduction paradigm. The extended phase space $\widetilde{M} = T^*\mathbb{R}^4$ has a natural S^1 symmetry whose momentum map level $\Xi=\,$const imposes the KS constraint [7]. Enforcing this constraint and reducing by S^1 yields the original Kepler phase space with a twisted symplectic form that encodes the S^1-bundle's curvature (Dirac monopole) [10]. This **2-body case** serves as a prototype: it demonstrates how adding an extra phase ("spinor") coordinate can trade a singular, non-integrable 3D problem for a regular, integrable higher-dimensional system, with **symplectic reduction** neatly linking the two [12] [7]. The next step (WP3–WP4) is to explore whether analogous *KS-like lifts* exist for the 3-body problem, and if so, whether their constraints can likewise be understood as momentum maps for certain symmetry groups. The hope, as evidenced by the Kepler case, is that the complex algebraic conditions of regularization might be reinterpreted as geometric moment-map conditions, revealing hidden symmetries and conserved quantities in the lifted space.

Sources: The above synthesis is supported by Van der Meer's reduction approach to KS [13] [7], Cushman's geometric analysis of the KS map [4] [6], and classical expositions of the Hopf fibration and monopole connection [11] [10], among others. The KS transformation was originally introduced in Kustaanheimo & Stiefel (1965) and elegantly reformulated via quaternions by Stiefel & Scheifele (1971). Modern treatments emphasize the bundle structure and symplectic reduction (e.g. Cordani 2003; Cushman & Bates 2015), solidifying the interpretation of KS constraints as moment-map conditions in the regularized Kepler problem. [8] [7]

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